

PRODUCT SPECIFICATION

Lithium Iron Phosphate
LFP (LiFePO₄) Battery

For (Battery Energy Storage System)

Product	LFP (LiFePO ₄) Battery
Model	RP_LFP-314
Spec.	3.2V / 314Ah
Date	2026-03-31 ~ 2026-12-31

Prepared	R&D Checked	QA Checked	Approved
/	/	/	/

Customer Approval	Date	Title / Name	Signature
	Company Name / Stamp		

Revision History

Ver.	Date	Mark	Revision Description	Originator
V1.0	2025-01-11	/	First Edition	R&D David
V1.1	2025-04-20	/	Change format	QA/QC Edward
V2.0	2025-07-01	/	Add battery test conditions	QA/QC Edward
V2.1	2025-10-01	/	Some changes to battery test conditions	QA/QC Edward
V2.2	2025-10-28	/	Some changes to battery test conditions	QA/QC Edward
V2.3	2025-11-05	/	Translation improvement	R&D Jin
V2.4	2025-12-10	/	Some changes to battery test conditions	QA/QC Edward
V2.5	2026-03-31	/	Added customer signature form and changed specification expression	QA/QC Edward

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PART 1.

Scope

- This specification is applied to the prismatic 314 Ah LFP battery cell of the square aluminum case supplied by Remplir Inc. Customer communicates with Remplir Inc. And if any special product specification is required, it should be agreed upon.

Model Name : RP_LFP-314

PART 2.

Definition of Terms

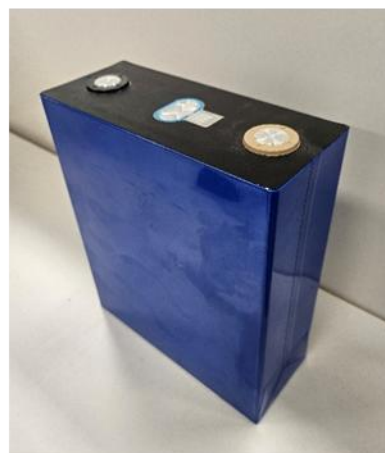
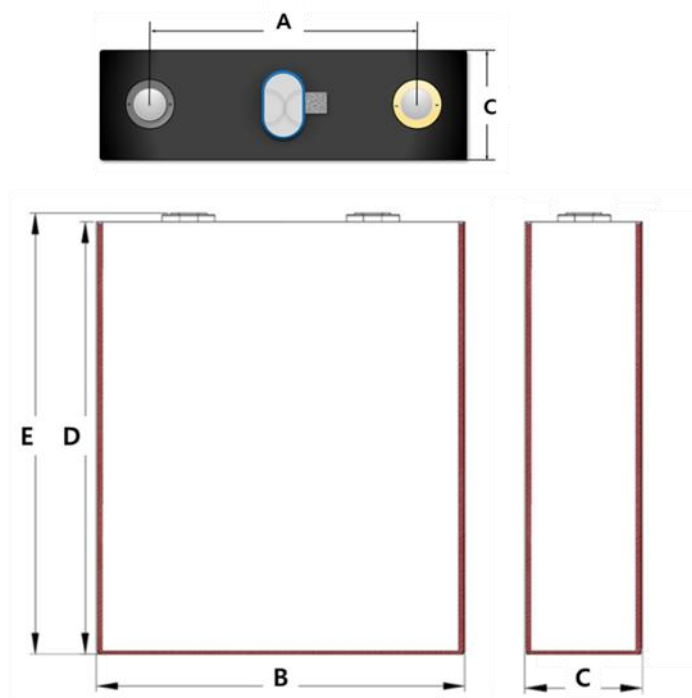
• Definition of Terms

No	Terms	Definition
2.1	Product	In this specification, "Product" refers to the prismatic-type 3.2V 314 Ah Lithium Iron Phosphate (LiFePO₄) battery cell manufactured by REMPILIR INC. (hereinafter, Remplir).
2.2	Ambient Temperature	This refers to the temperature of the surrounding environment where the battery cells are operated or located.
2.3	Cell Temperature	Temperature measured by a temperature sensor installed at the center of the cell's surface. The selection of the temperature sensor and the measurement line is to be jointly decided by Remplir and the customer.
2.4	C-Rate	The ratio of the charge/discharge current to the cell's capacity. For example, if the cell capacity is 314 Ah and the charge/discharge current is 157 A, the charge/discharge rate is 0.5 C. If the cell capacity decreases to 300 Ah and the charge/discharge current is 150 A, the charge/discharge rate is still 0.5 C.
2.5	State of Charge (SOC)	The ratio of the cell's capacity to its nominal capacity under no-load conditions. It is measured in Ah or Wh and indicates the cell's charging status. For example, if 314 Ah cell has 314 Ah / 157 Ah / 0 Ah capacity, SOC will be 100 % / 50 % / 0 % SOC, respectively.
2.6	Cycle	The process of a single charging and discharging loop according to specified charge/discharge criteria.
2.7	State of Health (SOH)	The ratio of the actual cell capacity to the nominal capacity after being used for a certain period. For example, a new cell with a nominal capacity of 314 Ah is considered to have 100 % SOH. If the cell capacity decreases to 251 Ah, it is considered to have 80 % SOH. A completely discarded cell has 0 % SOH.
2.8	Open Circuit Voltage (OCV)	This is the cell voltage measured under no-load conditions or when the circuit is disconnected.
2.9	Depth of Discharge (DOD)	This refers to the ratio of the cell's discharged capacity to its nominal capacity.
2.10	Customer	This refers to the buyer signed in the sales contract with Remplir Inc.

PART 3.

Product Dimensions

- Product Dimension (mm):



List	Description	Dimension
A	Distance between pole centers	123.0 ± 0.3 mm
B	Width	174.5 ± 0.5 mm
C	Thickness (100 ± 10 kgf)	71.7 ± 0.5 mm
	Thickness (Free)	71.7 ± 0.5 mm
D	Height (Shoulder)	204.2 ± 0.5 mm
E	Height (with terminal)	207.0 ± 0.5 mm
Weight	Weight	5.55 ± 0.15 kg

*The Battery pole is composed of dual-layered structure.

*Laser welding is recommended for connecting and assembling corresponding module/pack.

*The image shown is for reference only.

PART 4.
Product Specification
● Summarized Product Specification

No.	Items	Specification	Remark
4.1	Nominal Capacity	314 Ah	Standard Charging Method Standard Discharging Method
4.2	Nominal Energy	1004.8 Wh	
4.3	Nominal Voltage	3.2 V	
4.4	Operation Voltage	2.5 — 3.65 V	Cell temperature $0\text{ }^{\circ}\text{C} < T \leq 60\text{ }^{\circ}\text{C}$
		2.0 — 3.65 V	$-20\text{ }^{\circ}\text{C} < T \leq 0\text{ }^{\circ}\text{C}$
4.5	Standard Charging Method	CC(Constant current) charging at 0.5 C to 3.65 V, then CV(constant voltage) charging at 3.65 V until charging current declines to $\leq 0.05\text{ C}$ and rest for 30min.	$25 \pm 2\text{ }^{\circ}\text{C}$ $0.5\text{ C} = 157\text{ A}$ $0.05\text{ C} = 15.7\text{ A}$ Averaged capacity after the continuous 3 cycles
4.6	Standard Discharging Method	CC Discharging at 0.5 C to 2.5 V, and rest for 30min.	
4.7	AC-IR Cell Impedance	$0.18 \pm 0.05\text{ m}\Omega$	Internal resistance is measured at 1.0 kHz of AC frequency at SOC 30 %
4.8	Max. Charge Current	157 A / (0.5 C)	$25 \pm 2\text{ }^{\circ}\text{C}$
4.9	Max. Discharge Current	157 A / (0.5 C)	
4.10	Cycle Life	$\geq 6000\text{ cycles} / (25\text{ }^{\circ}\text{C})$	100 % DOD , 80 % SOH, Charge / Discharge (0.5C / 0.5C),
4.11	Operation Temperature Range	Charging: $0\text{ }^{\circ}\text{C} - 60\text{ }^{\circ}\text{C}$ Discharging: $-20\text{ }^{\circ}\text{C} - 60\text{ }^{\circ}\text{C}$	Cell temperature during the operation
4.12	Operation Humidity Range	10 — 95 % RH (Relative humidity)	Without precipitation
4.13	Shipment Capacity	$30 \pm 5\text{ \% SOC}$	Within 15 days from factory
4.14	Storage Temp	1 month : $-20 \sim 40^{\circ}\text{C}$ 3 month : $-10 \sim 35^{\circ}\text{C}$ 1 year : $25 \pm 2^{\circ}\text{C}$	The battery should be charged in 3 months, keep its charge state in about 50%
4.15	Structural Stability	No stain, No rupture, No deformation, No leakage-	
4.16	Standard Testing Condition	Temperature: $25 \pm 2\text{ }^{\circ}\text{C}$ / Humidity: $\leq 80\text{ \% RH}$ (Relative humidity) Atmospheric pressure: 86 — 106 kPa	

PART 5.**Agreements**

Any other matters not included in this specification must be agreed between the manufacturer (Remplir Inc.) and the customer.

5.1 Any applications or operating conditions that differ from those specified in this document **must be discussed and approved** in advance by the manufacturer (Remplir Inc.). Additional testing may be required to verify the performance and safety of the product under such conditions.

5.2 The manufacturer shall not be held responsible for any accidents, damages, or malfunctions resulting from the use of the battery under conditions **not specified in, or not approved in writing by**, this document.

5.3 Any matters not covered in this specification shall be determined **through mutual agreement** between the manufacturer and the customer.

5.4 This product complies with the <Eco-Friendly and Hazardous Substance Management Standards for Manufactured Products>.

5.5 Manufacturer **reserves the right** to revise product specifications. The customer **must verify the latest version** of the product specification prior to purchase.

5.6 In the event of any discrepancy or inconsistency between the English and Korean versions of this document, **the Korean version shall prevail.**

PART 6.**Manufacturer Information**

[Company : Remplir Inc.]

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